

NECEDAH 5 WNW, WI, USA

WMO: 743580

Lat: 44.0604N

Lon: 90.1737W

Elev: 933

StdP: 14.21

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54903

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-6.5	-17.7	2.2	-9.9	-13.3	2.8	-5.6	19.4	13.1	17.5	15.8	4.3	N/A	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	88.1	73.2	85.4	71.3	82.7	69.7	75.7	84.9	73.9	82.2	72.2	79.7	7.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	126.8	80.6	71.1	118.8	78.6	69.5	112.1	77.2	39.8	85.0	38.0	82.6	36.5	79.8	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	14.7	12.8	DB	-20.1	92.2	6.7	3.0	-24.9	94.4	-28.8	96.1	-32.5	97.8	-37.3	99.9
			WB	-18.5	78.9	5.9	2.0	-22.8	80.3	-26.2	81.5	-29.5	82.6	-33.8	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.7	16.0	19.2	32.8	46.0	58.2	67.2	71.0	68.5	61.1	48.6	35.5	22.3		
	DBStd	21.45	12.61	12.12	12.58	9.64	8.82	5.86	5.59	5.34	8.07	9.11	10.01	11.28		
	HDD50	4130	1053	862	553	187	24	0	0	0	9	140	444	858		
	HDD65	7590	1518	1282	1000	572	247	42	13	25	167	516	885	1323		
	CDD50	2556	0	0	19	66	280	517	652	575	343	95	9	0		
	CDD65	541	0	0	2	2	37	109	199	135	51	6	0	0		
	CDH74	4646	0	0	12	42	389	900	1779	1037	431	56	0	0		
	CDH80	1227	0	0	0	5	98	220	561	239	98	6	0	0		
	WSAvg	5.6	6.2	6.3	6.3	7.3	6.3	5.1	4.4	3.9	4.3	5.3	5.9	5.9		
	PrecAvg	32.2	1.0	0.8	1.6	3.0	3.9	5.1	4.4	4.2	3.3	2.3	1.6	1.2		
(15) Precipitation	PrecMax	42.2	2.4	1.8	3.3	6.1	9.3	10.4	13.0	8.8	8.7	5.5	4.4	2.6		
	PrecMin	23.3	0.4	0.1	0.1	0.8	1.6	1.5	1.3	1.1	0.8	0.5	0.0	0.2		
	PrecStd	4.8	0.5	0.4	0.9	1.3	1.6	2.7	2.9	1.9	1.9	1.3	1.0	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	55.5	75.5	79.5	88.2	89.6	93.9	89.1	87.3	80.3	67.7	50.6		
		MCWB	39.6	48.9	60.6	59.4	68.7	74.3	75.9	73.7	72.6	62.6	54.1	46.8		
	2%	DB	41.3	45.7	64.2	72.6	82.4	85.7	89.2	85.5	83.0	74.1	60.8	44.5		
		MCWB	37.2	41.1	53.6	57.4	64.8	70.5	75.2	72.0	70.5	61.0	51.7	40.7		
	5%	DB	36.9	40.3	57.7	67.8	78.3	82.3	85.8	82.6	78.9	69.3	55.4	40.6		
		MCWB	33.9	35.8	48.6	52.8	62.8	68.2	72.3	70.7	67.2	59.4	47.5	37.5		
	10%	DB	33.6	35.6	50.6	62.7	74.2	79.3	82.8	79.9	75.5	64.1	50.7	36.9		
		MCWB	31.8	31.9	43.2	49.8	61.3	67.0	70.8	68.8	65.6	56.8	44.4	34.3		
	0.4%	WB	40.7	47.2	62.2	62.4	72.2	76.7	79.6	76.4	74.7	67.6	57.0	49.2		
		MCDB	44.9	54.7	74.1	73.6	84.2	86.0	88.7	84.5	84.2	71.3	62.7	50.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	41.8	57.7	58.4	68.8	73.4	76.7	74.6	72.2	63.4	52.6	41.9		
		MCDB	40.9	46.6	63.4	69.8	79.0	82.8	85.8	82.3	80.2	68.4	60.8	43.2		
	5%	WB	34.4	36.5	49.6	54.7	66.5	71.1	74.7	72.7	69.9	60.1	48.7	37.8		
		MCDB	36.4	40.4	56.4	64.1	75.4	78.7	83.6	79.7	75.4	67.4	54.5	40.8		
	10%	WB	32.0	32.8	44.1	50.5	63.7	68.7	72.8	70.9	67.6	56.5	44.7	34.7		
		MCDB	34.1	36.1	51.3	60.6	72.0	76.4	80.2	77.8	73.1	62.8	49.4	37.1		
	5% DB	MDBR	16.4	17.7	18.6	19.0	19.9	18.9	20.0	19.5	20.8	18.4	15.5	13.8		
		MCDBR	18.1	20.4	25.2	28.0	25.5	22.6	22.8	21.6	23.7	24.2	23.3	17.5		
		MCWBR	14.8	14.7	14.4	14.2	11.6	10.7	9.9	9.7	11.7	14.0	15.1	13.7		
		MCWBR	16.6	19.5	21.1	23.9	21.3	19.1	19.5	18.4	19.0	21.4	22.3	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.290	0.323	0.358	0.389	0.399	0.393	0.397	0.362	0.325	0.290	0.265		
		taud	2.399	2.309	2.378	2.355	2.315	2.316	2.341	2.331	2.423	2.516	2.529	2.452		
	Ebn at Noon		279	286	291	287	280	276	277	272	274	272	266	266		
		Edh at Noon	23	31	33	37	40	40	39	38	32	25	21	20		
(44) All-Sky Solar Radiation	RadAvg		552	903	1243	1432	1709	1884	1939	1655	1325	845	546	406		
	RadStd		64	82	146	136	131	143	96	111	100	102	50	44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page